Example 4.54 From a class of 25 students, 10 students are to be chosen for an excursion party. There are 4 students who decide that either all of them will join or none of them will join. In how many ways can the excursion party be chosen?

Solution:
There are two possibilities (i) All the 4 students will got to the excursion party then, we need to select 6 students out of 21 students. It can be done in ${}^{21}C_6 = \dfrac{21!}{6! \times 15!}$ ways.

(ii) All the 4 students will not go to the excursion party then, we need to select 10 students out of 21 students. It can be done in ${}^{21}C_{10} = \dfrac{21!}{10! \times 11!}$. Hence, the total number of ways is ${}^{21}C_6 + {}^{21}C_{10} = \dfrac{21!}{6! \times 15!} + \dfrac{21!}{10! \times 11!}$.

Example 4.55 A box of one dozen apple contains a rotten apple. If we are choosing 3 apples simultaneously, in how many ways, one can get only good apples.

Solution:
The total number of ways of selecting 3 apples from 12 apples is ${}^{12}C_3 = 220$.

The total number of ways of getting a rotten apple when selecting 3 apples from 12 apples is equal to selecting 1 rotten apple and remaining 2 apples can be selected from 11 apples is ${}^{11}C_2 = 55$.

Therefore, the total number of ways of getting only good apples is

$$ {}^{12}C_3 - {}^{11}C_2 = 220 - 55 = 165$$

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Example 4.56 An exam paper contains 8 questions, 4 in Part A and 4 in Part B. Examiners are required to answer 5 questions. In how many ways can this be done if
(i) There are no restrictions of choosing a number of questions in either parts.
(ii) At least two questions from Part A must be answered.

Solution:

(i) **There are no restrictions.** Totally there are 8 questions in both Part A and Part B. The total number of ways of attempting 5 questions from 8 questions is ${}^8C_5 = {}^8C_3 = 56$.

(ii) **At least two questions from Part A needs to be answered.**Accordingly, various choices are tabulated as follows.

Part A	Part B	Number of selections
2	3	${}^4C_2 \times {}^4C_3$
3	2	${}^4C_3 \times {}^4C_2$
4	1	${}^4C_4 \times {}^4C_1$

Therefore, the required number of ways of answering is

$$ {}^4C_2 \times {}^4C_3 + {}^4C_3 \times {}^4C_2 + {}^4C_4 \times {}^4C_1 = 24 + 24 + 4 = 52.$$